

WTW 158 Worksheet 6 on Limits at Infinity and Differentiability

A. TEXTBOOK PROBLEMS

p137 no 3, 17, 19, 22, 27, 33, 35, 37, 42, 49

p148 no 3, 20, 22, 37

p160 no 3, 4, 41

B. MORE PROBLEMS

True or false? Explain by quoting a theorem or by giving an example that shows it is false.

(i) If f is continuous at a then f is differentiable at a .

(ii) A function can only have one horizontal asymptote.

(iii) A function cannot intersect a horizontal asymptote.

(iv) A function can intersect a vertical asymptote.

(v) If $|f|$ is continuous at a then f is continuous at a .

(vi) If $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow \infty} g(x) = \infty$ then $\lim_{x \rightarrow \infty} (f(x) - g(x)) = 0$.

(vii) If $f(1) > 0$ and $f(3) < 0$ then there exists a number c between 1 and 3 such that $f(c) = 0$.

(viii) If $f'(a)$ exists then $\lim_{x \rightarrow a} f(x) = f(a)$.

C. TEST QUESTIONS

Question 1

The largest interval on which the function $f(x) = \sqrt{x+4}$ is differentiable, is

(a) $[0, \infty)$	(b) $(0, \infty)$	(c) $[-4, \infty)$	(d) $(-4, \infty)$	(e) $(-\infty, \infty)$
(f) None of these				

Question 2

The equation(s) of the horizontal asymptote(s) of $f(x) = \frac{e^x}{1+e^x}$ is(are)

(a) $y = -1$ and $y = 1$	(b) $x = 1$	(c) $y = 0$ and $y = 1$	(d) $y = 0$
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Question 3

If $f(x) = \begin{cases} -1 & \text{if } x \leq 0 \\ e^{-x} & \text{if } x > 0 \end{cases}$, then f is

(a) continuous and differentiable at 0
(b) continuous but not differentiable at 0
(c) not continuous but differentiable at 0
(d) not continuous and not differentiable at 0

D. YOUTUBE VIDEOS

Limits at infinity - Basic ideas and shortcuts - Patrickjmt

Differentiability - Elizabeth Stepp

E. SELECTED ANSWERS

A. p137 3(a) -2 (b) 2 (c) ∞ (d) $-\infty$ (e) HA: $y = -2, y = 2$ VA: $x = 1, x = 3$ 17. $-\frac{1}{2}$
19. -1 22. -3 33. $-\infty$ 35. $\frac{\pi}{2}$ 37. $-\frac{1}{2}$ 42. 0 49. $y = 2$ is a HA.

p148 3. (i) 2 (ii) 2 20. $y = 4x - 23$ 22. $f(4) = 3, f'(4) = \frac{1}{4}$ 37. $f(x) = \sqrt{x}, a = 9$.

p160 3. (a) II (b) IV (c) I (d) III 41. f is not db in $x = -4$ because left and right derivatives are not the same. f is not db in $x = 0$ because f is discontinuous in $x = 0$.

B. (i) - (vii) F (viii) T

C. 1(d) 2(c) 3(d)

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p 137

3. (a) -2 (b) 2 (c) 1 (d) -∞

(e) HA: $y = -2$, $y = 2$ VA: $x = 1$, $x = 3$

$$\begin{aligned} 17. \lim_{x \rightarrow -\infty} \frac{1-x-x^2}{2x^2-7} \\ = \lim_{x \rightarrow -\infty} \frac{\frac{1}{x^2} - \frac{1}{x} - 1}{2 - \frac{7}{x^2}} \\ = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} 19. \lim_{t \rightarrow \infty} \frac{\sqrt{t} + t^2}{2t - t^2} \\ = \lim_{t \rightarrow \infty} \frac{\frac{1}{t^{3/2}} + 1}{\frac{2}{t} - 1} \\ = -1 \end{aligned}$$

$$\begin{aligned} 22. \lim_{x \rightarrow -\infty} \frac{\sqrt{9x^6 - x^7}}{x^3 + 1} \\ = \lim_{x \rightarrow -\infty} \frac{|x^3| \sqrt{9 - \frac{1}{x^5}}}{1 + \frac{1}{x^3}} \\ = \lim_{x \rightarrow -\infty} \frac{-x^3 \sqrt{9 - \frac{1}{x^5}}}{1 + \frac{1}{x^3}} \\ = -3 \end{aligned}$$

$$\begin{aligned} 33. \lim_{x \rightarrow -\infty} (x^2 + 2x^3) \\ = \lim_{x \rightarrow -\infty} x^3 \left(\frac{1}{x^5} + 2 \right) \\ = -\infty \end{aligned}$$

$$\begin{aligned} 35. \lim_{x \rightarrow \infty} \arctan(e^x) \\ = \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} 37. \lim_{x \rightarrow \infty} \frac{1-e^x}{1+2e^x} \\ = \lim_{x \rightarrow \infty} \frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 2} = -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} 42. \lim_{x \rightarrow \infty} [\ln(2+x) - \ln(1+x)] \\ = \lim_{x \rightarrow \infty} \ln\left(\frac{2+x}{1+x}\right) \\ = \lim_{x \rightarrow \infty} \ln\left(\frac{\frac{2}{x} + 1}{\frac{1}{x} + 1}\right) \\ = \ln 1 = 0 \end{aligned}$$

$$\begin{aligned} 49. \lim_{x \rightarrow \pm\infty} \frac{2x^2 + x - 1}{x^2 + x - 2} \\ = \lim_{x \rightarrow \pm\infty} \frac{2 + \frac{1}{x} - \frac{1}{x^2}}{1 + \frac{1}{x} - \frac{2}{x^2}} \\ = 2 \end{aligned}$$

$\Rightarrow y = 2$ is a HA
 $x = 1, -2$ are VA

p 148

$$\begin{aligned} 3. (i) x=1: \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x-1} \\ = \lim_{x \rightarrow 1} \frac{4x - x^2 - 3}{x-1} \\ = \lim_{x \rightarrow 1} \frac{-1}{-x+1} \frac{(x-3)}{x-1} \\ = \lim_{x \rightarrow 1} \frac{-1}{x-1} (x-3) \\ = \lim_{x \rightarrow 1} -x + 3 = 2 \end{aligned}$$

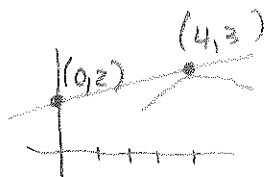
$$\begin{aligned} (ii) \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ = \lim_{h \rightarrow 0} \frac{4(1+h) - (1+h)^2 - 3}{h} \\ = \lim_{h \rightarrow 0} \frac{4 + 4h - 1 - 2h - h^2 - 3}{h} \\ = \lim_{h \rightarrow 0} \frac{-h^2 + 2h}{h} \\ = \lim_{h \rightarrow 0} (-h + 2) = 2 \end{aligned}$$

20. $y+3 = 4(x-5)$

$\Rightarrow y = 4x - 23$

22. $f(4) = 3$

$f'(4) = \frac{3-2}{4-0} = \frac{1}{4}$



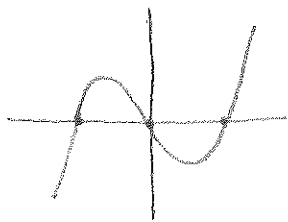
37. $f(x) = \sqrt{x}$, $a = 9$

p160

3. (a) - II (b) - IV

(c) - I (d) - III

4



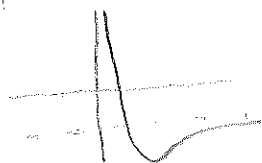
41. f is not db in $x = -4$ because left & right derivatives are not the same.

f is not db in $x = 0$ because f is discontinuous in $x = 0$

B. (i) False. If f is db in a then f is continuous in a .

(ii) False. $y = \arctan x$ has 2 HAs

(iii) False. Eg.

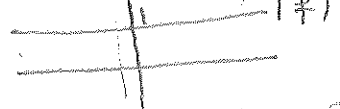
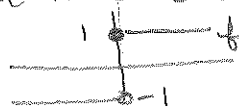


(iv) False. It is no longer a function if that is the case.

(v) False. For

$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$

For this function $|f|$ is continuous at $x = 0$ but f is not continuous.



(vi) False. Let $f(x) = x$, $g(x) = x^2$
Then $\lim_{x \rightarrow \infty} f(x) = \infty$ & $\lim_{x \rightarrow \infty} g(x) = \infty$

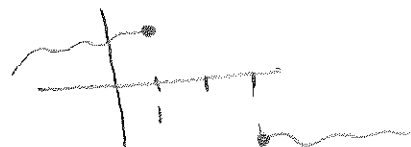
But $\lim_{x \rightarrow \infty} (f(x) - g(x))$

$= \lim_{x \rightarrow \infty} (x - x^2)$

$= \lim_{x \rightarrow \infty} x(1 - x)$

$= -\infty$

(vii) False.



f could be discontin.

(viii) Yes, differentiability implies continuity

Q1. (d)

(2) (c)

(3) (d)